Here, we focus on the feasibility of detecting non-thermal bremsstrahlung emission from the Type Ia model W7 at t = 40 days after the explosion. The calculated flux at t = 40 days from Clayton & The's Figure 13 at a distance of 1 Mpc and The & Burrows 1994 Figure 3 of SN 2014J distance of 3.5 Mpc are shown below.

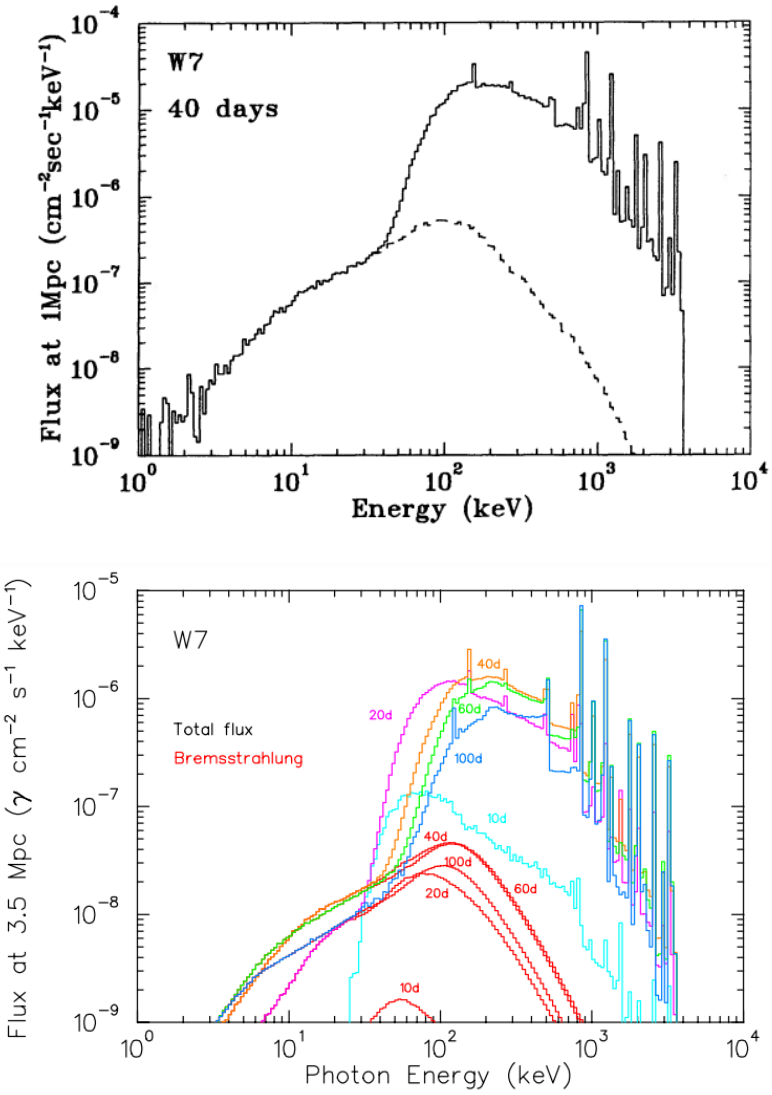


Figure 3. Flux spectra at a distance of 3.5 Mpc in photons cm^{-2} s^{-1} keV^{-1} for the emergent hard X-ray photons and gamma-ray lines 10, 20, 40, 60, and 100 days after explosion for the fiducial model W7. The photon energy is in keV from 1 keV to ∼3.2 MeV. Included are curves for the total flux and its bremsstrahlung contribution, indicating the importance of bremsstrahlung below ∼60 keV. Note that the line fluxes depicted are per plotting bin width, and are not resolved. They show the line positions, but the relative heights are only crude measures of the actual fluxes, whose integrals (the total line fluxes) are shown correctly in photons cm^{-2} s^{-1} in Figures 14–16.

The figures show that the flux at 10 keV at a distance of 1 Mpc is ~5e-8 ph/cm^2/s/keV and at a distance of 3.5 Mpc is ~6e-9 ph/cm^2/s/keV.

Below is the 3 sigma sensitivity of X-ray and gamma-ray detectors from arXiv:1709.10414v1 [astro-ph.IM] 29 Sep 2017 (Cavallari & Frontera 2017).

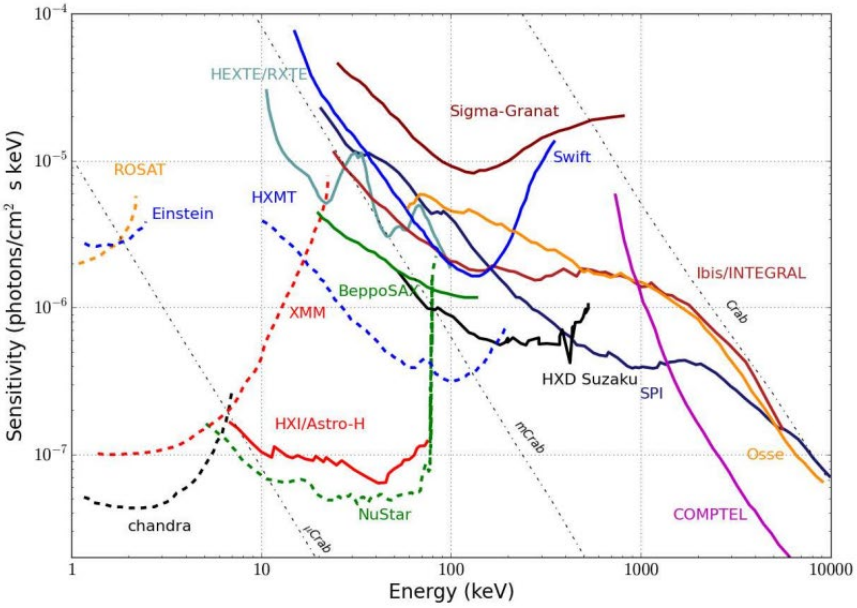


Fig. 21 Three sigma sensitivity, for an exposure time $T = 10^5$ s and a bandwidth $\Delta E = E/2$, of the most relevant missions in the hard X–ray/soft gamma–ray band. Adapted from Virgilli et al. (2017).

The most relevant detector is NuStar, which has a 3sigma sensitivity of 6e-8 ph/cm^2/s/keV at 10 keV. Therefore, if a W7 supernova explodes at a distance of 1 Mpc, NuStar could detect it. And we need the next generation X-ray detector of a magnitude more sensitive than NuStar to detect a W7 supernova at a distance of 3.5 Mpc (the distance of the M82 galaxy where SN2014J is located).

The most promising in detecting the non-thermal bremsstrahlung from Type Ia supernovae is the proposed HEX-P mission.

Note that the 847 keV and the 1238 keV gamma-line fluxes from SN2014J were detected by the INTEGRAL detector. Also, the 70-190 keV and the 200-400 keV continuum of SN2014J were detected by the INTEGRAL detector (compare the fluxes in the Figure 3 and Fig. 21 above).

As of early 2026, the **High Energy X-ray Probe (HEX-P)** remains a cornerstone of the next generation of X-ray astronomy, specifically designed to bridge the gap between soft X-rays and gamma-rays that will be left when **NuSTAR** eventually de-orbits.

Latest Mission Status & Context (2025–2026)

- **Mission Class:** HEX-P is a **NASA Probe-class** mission concept. It was designed to answer the call of the *Astro2020 Decadal Survey* for a sensitive, broad-band X-ray observatory to serve the 2030s.

- **Competitive Landscape:** In a significant development for the X-ray community in March 2026, a major "competitor" mission concept, **AXIS** (Advanced X-ray Imaging Satellite), was reportedly ruled ineligible for further selection due to administrative and budgetary complications. This potentially positions HEX-P as the primary candidate for the next major X-ray probe.

- **Scientific Urgency:** With flagship missions like **Chandra** aging and the ESA's **Athena** not scheduled until at least 2037, HEX-P is being championed as the essential "Guest Observer" facility to prevent a "discovery gap" in the 2030s.

Key Technical Breakthroughs

Recent publications (Frontiers in Astronomy, 2024–2025) highlight the specific technological leaps that enable the sensitivity study you were modeling:

- **Effective Area:** HEX-P achieves an effective area **superior to NuSTAR** above 10 keV through advanced Pt/C and W/Si multilayer coatings on its mirrors.

- **Dual-Telescope Suite:**
    - **High Energy Telescope (HET):** Covers **2–80 keV** using two mirror assemblies.
    - **Low Energy Telescope (LET):** Covers **0.2–20 keV** using silicon segmented mirror technology for high spatial resolution.

- **Observing Efficiency:** Designed for the **L1 Lagrange point**, HEX-P will have $>90\%$ observing efficiency, allowing for the long, uninterrupted exposures (like your 100 ks example) required to catch faint SNe Ia signals.

Recent Science Focus: Supernovae & Nuclear Physics

Scientific articles published in late 2024 and 2025 have refined the case for HEX-P in the realm of supernovae:

- **Type Ia SNe:** Simulations confirm HEX-P can detect the hard X-ray continuum (bremsstrahlung) from SNe Ia out to the **Virgo Cluster (~20 Mpc)**, providing a far larger statistical sample than NuSTAR's current limit (~ 5 Mpc).

- **44Ti Detection:** HEX-P is expected to map the distribution of **radioactive ^{44}Ti** in young supernova remnants with unprecedented detail, helping to "look inside" the explosion engine.

- **Nuclear Astrophysics:** It will be sensitive enough to search for rare nuclear lines, potentially identifying sites of the **r-process** (where heavy elements like gold are made) in neutron star mergers and specific supernovae.

Recommended Reading

1. **"The High Energy X-ray Probe (HEX-P): Science Overview"** (*Frontiers in Astronomy and Space Sciences*, 2024) – Best for understanding the broad mission goals.
2. **"Instrument and Mission Profile"** (*NASA Technical Reports Server*, 2024) – Best for the technical specs needed for sensitivity modeling.
3. **"Supernova Remnants, Pulsar Wind Nebulae, and Nuclear Astrophysics"** (*Frontiers*, 2023) – The specific science case for your supernovae study.

This summary is designed to be a core narrative for a science case, emphasizing why **HEX-P** is a transformative upgrade over **NuSTAR** for Type Ia supernova (SNe Ia) research.

Science Case: Resolving the SNe Ia Engine with HEX-P

1. The Scientific Objective

Despite their use as standard candles, the progenitor systems of Type Ia supernovae remain a central mystery in astrophysics. The hard X-ray (>10 keV) emission produced by **bremsstrahlung from energetic secondary electrons** (born from Compton-scattered gamma rays) provides a "clean" look at the total mass and density structure of the ejecta, as this radiation is significantly less affected by photoelectric absorption than soft X-rays.

2. The NuSTAR Constraint (The "Local Only" Problem)

Currently, our ability to observe this emission is limited to extremely nearby events.

- **SN 2014J** (at ~ 3.5 Mpc) provided a landmark detection, but even then, the signal was near the instrumental noise floor.
- At this distance limit, the sample size of SNe Ia is statistically insignificant (roughly 1 event per decade).

3. The HEX-P Solution: Accessing the Virgo Cluster

HEX-P expands the discovery horizon by a factor of $6\times$ in distance, meaning it can reach the **Virgo Cluster (~16.5 Mpc)**.

Metric	Scientific Impact
Volume Search	Reaching Virgo increases the available galaxy count by $\sim 200\times$, enabling the study of multiple SNe Ia annually.
Spectral Fidelity	Improved angular resolution ($<10"$) suppresses the background, allowing for the first high-precision measurements of the $20-60$ keV continuum slope.
Density Probing	By measuring the timing of when the ejecta become transparent to bremsstrahlung, HEX-P can distinguish between **Single Degenerate** (accreting white dwarf) and **Double Degenerate** (merger) models.

4. Direct Progenitor Discrimination

Because the bremsstrahlung flux is sensitive to the **total mass of the ejecta**, HEX-P can directly test whether "sub-Chandrasekhar" mergers or "standard" Chandrasekhar-mass explosions are responsible for the diversity of SNe Ia. NuSTAR simply lacks the sensitivity to detect these differences at distances beyond the immediate Local Group.

Summary for Proposal

"HEX-P will transition SNe Ia hard X-ray studies from an era of 'lucky detections' of rare nearby events to a systematic population study. By reaching the Virgo Cluster with a 3\sigma sensitivity an order of magnitude better than NuSTAR, HEX-P will provide the definitive observational constraints needed to resolve the decades-old progenitor debate."